

Statistical and Computational Models

Lecture 2: Linear regression as statistical learning

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Roadmap

Linear regression redux

RSS and the EPE

Two choices

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Loss: What kind of 'error' matters, and how we price those errors.

Today we pick one of each. The linear model combines:

- Functions of the form $\mathbf{x}^\top \boldsymbol{\beta}$
- Mean squared error loss

The design matrix

Returning to our voting example, linear regression requires an $n \times 1$ **response vector**, an $n \times (p + 1)$ **design matrix** whose first column is ones, and a $(p + 1) \times 1$ **coefficient vector**:

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{\mathbf{y}} \quad \underbrace{\begin{bmatrix} 1 & \text{age}_1 & \text{mail}_1 \\ 1 & \text{age}_2 & \text{mail}_2 \\ \vdots & \vdots & \vdots \\ 1 & \text{age}_n & \text{mail}_n \end{bmatrix}}_{\mathbf{X}} \quad \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}}_{\boldsymbol{\beta}}$$

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So $\mathbf{X}\boldsymbol{\beta}$ is $n \times 1$ – one fitted value per cell.

Question: what does the column of ones do? Allows the mean of $\mathbf{Y}$ to be non-zero when all covariates are zero.

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Now turn that loss into an **objective function** – sum it over the sample:

$$\text{RSS}(\mathbf{b}) = \sum_{i=1}^n \left(y_i - \mathbf{x}_i^\top \mathbf{b} \right)^2 = (\mathbf{y} - \mathbf{X}\mathbf{b})^\top (\mathbf{y} - \mathbf{X}\mathbf{b})$$

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Read: find the $\mathbf{b}$ that minimises the objective, and so the summed loss.

Solving for $\hat{\beta}$

Step 1: expand the objective.

$$\text{RSS}(\mathbf{b}) = \mathbf{y}^\top \mathbf{y} - 2\mathbf{b}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{b}^\top \mathbf{X}^\top \mathbf{X} \mathbf{b}$$

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Step 2: differentiate with respect to $\mathbf{b}$, and set to zero.

$$-2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{0}, \quad \text{so} \quad \mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{y}$$

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Step 3: simplify, which requires $\mathbf{X}$ of **full column rank**.

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

QED

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2. Regress **y** on **Z**. Keep the residuals, $\tilde{\mathbf{y}}$.
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2. Regress **y** on **Z**. Keep the residuals, $\tilde{y}$.
3. Regress $\tilde{y}$ on $\tilde{x}$.

Read: strip from both age and turnout everything **Z** can explain, then relate what is left over.

Partial interpretation

```
xt <- resid(lm(x ~ mail, data = train)) # age, mail projected out
yt <- resid(lm(y ~ mail, data = train)) # turnout, mail projected out

c(residualised = coef(lm(yt ~ xt))[["xt"]],
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Residualising y is actually optional. $\tilde{x}$ is already orthogonal to Z , so whatever Z explains cannot reach $\hat{\beta}_{\text{age}}$.

Question: what if age is nearly a linear function of Z ?

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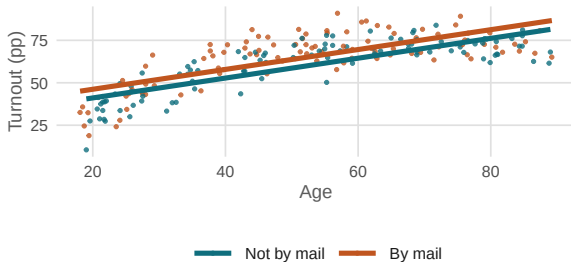
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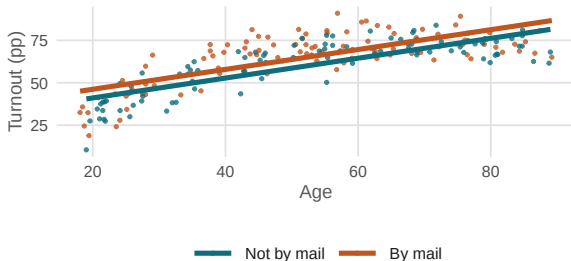
Question: what if age is nearly a linear function of $\mathbf{Z}$? Then $\tilde{\mathbf{x}}$ is tiny, almost no variation in age survives, and $\hat{\beta}_{\text{age}}$ is mostly noise. This is 'multicollinearity.'

Partial interpretation



Raw cells, split by **Z**. One fit per group.

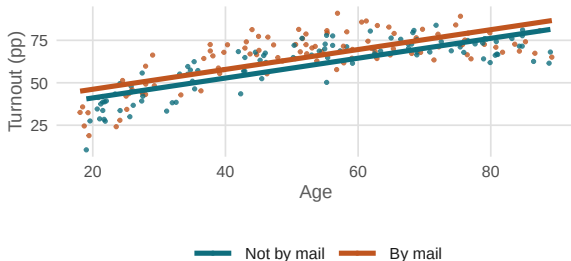
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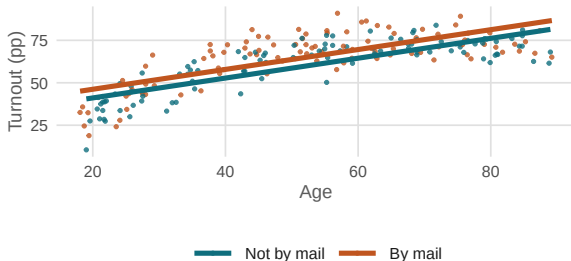


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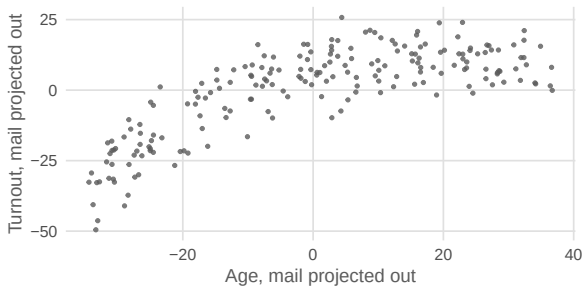


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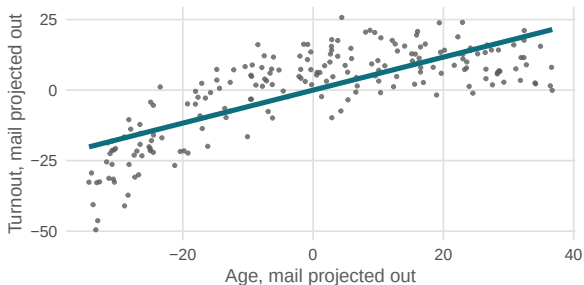
Question: what does partialling out **Z** remove? The gap between them.

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Each point is one cell, with **Z** projected out of **both** axes (residualised).

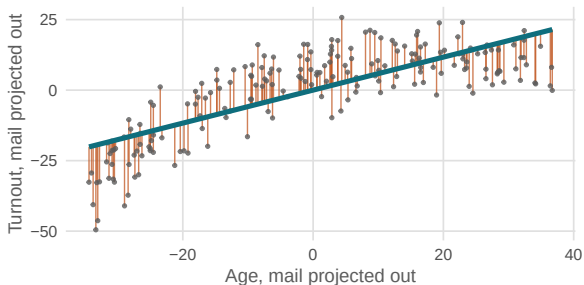
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The slope of that line is $\hat{\beta}_{\text{age}}$ – the number the multiple regression already gave us.

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The slope of that line is $\hat{\beta}_{\text{age}}$ – the number the multiple regression already gave us.

It is the line minimising the sum of the squared **residuals** (the vertical drops).

Roadmap

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RSS and the EPE

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Read: $f^\dagger$ is the best function **in the class**, and $f^\dagger$ need not be f^* .

Note: The loss still fixes the **kind** of answer we get.

Three functions

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$$f^* \longrightarrow f^\dagger \longrightarrow \hat{f}$$

Read: OLS is two steps removed from f^* . The first comes from restricting the **class**, the second from using a **sample**.

The best linear predictor

$f^\dagger$ is linear, so a coefficient vector pins it down. Define β^* as the one minimising $\mathbb{E}[(Y - \mathbf{X}^\top \beta)^2]$. Solving that gives:

$$f^\dagger(\mathbf{x}) = \mathbf{x}^\top \beta^*, \quad \beta^* = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]^{-1} \mathbb{E}[\mathbf{X}Y]$$

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Note: β^* exists whether or not $\mathbb{E}[Y | \mathbf{X}]$ is a straight line (in this case it is not).

Training vs. test vs. EPE

Lecture 1 ended with this table:

Quantity	Averages the loss over	Observable?
Training loss $\frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}(\mathbf{x}_i))$	the n observations we fit $\hat{f}$ on	always
Test loss $\frac{1}{m} \sum_{j=1}^m L(y_j, \hat{f}(\mathbf{x}_j))$	m observations not used in fitting $\hat{f}$	only if you have unseen data
EPE $\mathbb{E}[L(Y, \hat{f}(\mathbf{X}))]$	the population (or very many draws from the DGP)	never

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Question: which row can we actually minimise? The first one.

Substitute the one we can compute:

$$\text{MSE}_{\text{train}}(f) = \frac{1}{n} \sum_{i=1}^n \left(y_i - f(\mathbf{x}_i) \right)^2, \quad \hat{f} = \arg \min_{f \in \mathcal{F}} \text{MSE}_{\text{train}}(f)$$

Read: replace the expectation with a sample average, and minimise that instead.

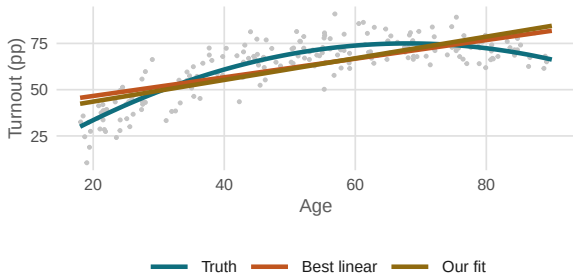
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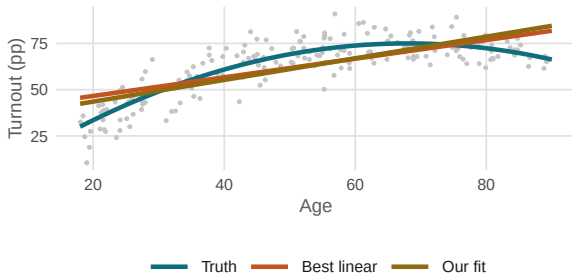
Read: replace the expectation with a sample average, and minimise that instead.

$\text{MSE}_{\text{train}}$ is just RSS/n . So minimising RSS is minimising an **estimate** of EPE.

Three functions

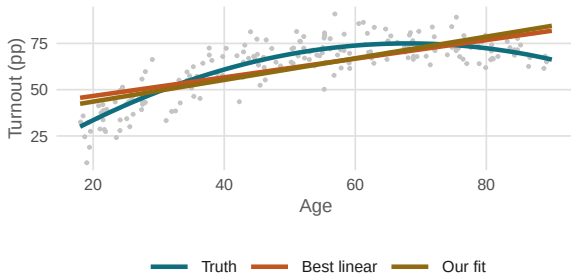


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Read: the gap between the first two comes from the class restriction. The gap between the last two comes from the sample.

Approximation and estimation

APPROXIMATION ERROR

$$\text{EPE}(f^\dagger) - \text{EPE}(f^\star)$$

From the **class** restriction.

Fixed once $\mathcal{F}$ is chosen, and will not shrink as $n \rightarrow \text{inf}$.

ESTIMATION ERROR

$$\text{EPE}(\hat{f}) - \text{EPE}(f^\dagger)$$

From **sampling**.

Not fixed, and shrinks as $n \rightarrow \text{inf}$.

In our case, the approximation error costs us **35.5** against a noise floor of **49**.

What consistency buys you

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Question: does this get us to f^* ? No! Approximation error may remain.

References

- James, G., Witten, D., Hastie, T. & Tibshirani, R. (2021) *An Introduction to Statistical Learning*, 2nd ed. – **2.2, 3.1–3.2**. statlearning.com.
- Efron, B. & Hastie, T. (2016) *Computer Age Statistical Inference*, ch. 7.

Start with 2.2. Sections 3.1–3.2 should already be familiar.